**KQC7031 INDUSTRIAL AUTOMATION & ROBOTICS**

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# ASSIGNMENT


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# Case I: Derivation of the Optimal Vaccination Control

*SIR Epidemic Model - Unconstrained (No ICU State Constraint)*

Della Rossa, Freddi & Goreac (2025). JOTA 204:8. DOI: 10.1007/s10957-024-02598-w

## Problem Statement

Minimise the cost functional:

$$J(v) = \int_0^T [\lambda_v \, v\,(t) + \lambda_i \, i\,(t)] \, \mathrm{dt}$$

subject to the SIR state equations:

$$\dot{s} = -\beta si - vs,$$
$$i = \beta si - \gamma i,$$

Parameters: $\beta = 0.30$, $\gamma = 0.10$, $\lambda_v = 2.0$, $\lambda_i = 15.0$, s(0) = 0.95, i(0) = 0.01, T = 50, $0 \le v(t) \le v_m = 0.05$. No state constraint on i(t) in Case I.

## Step 1 : Form the Hamiltonian

The Hamiltonian for a minimisation problem is $H = L + p_s s_t + p_i f_i$, where:

$$L = \lambda_v v + \lambda_i i$$
$$f_s = -\beta si - vs$$
$$f_i = \beta si - \gamma i$$

*Substituting into H:*

$$H = (\lambda_v v + \lambda_i i) + p_s(-\beta si - vs) + p_i(\beta si - \gamma i)$$

*Distributing each bracket:*

$$p_s(-\beta si - vs) = p_s \cdot -\beta si + p_s \cdot -vs = -\beta si p_s - vs p_s$$
$$p_i(\beta si - \gamma i) = p_i \cdot \beta si + p_i \cdot -\gamma i = \beta si p_i - \gamma i p_i$$

*Collecting all terms:*

$$H = \lambda_v v + \lambda_i i - \beta si p_s - vs p_s + \beta si p_i - \gamma i p_i$$

*Grouping terms that multiply v:*

$$H = (\lambda_v - p_s s)v + \lambda_i i + \beta si(p_i - p_s) - \gamma p_i i$$

**Step 2 : Find the Co-State Equation for $p_s$s**

PMP co-state rule:

$$\dot{p}_s = -\frac{\partial H}{\partial s}$$

*Differentiating H term by term with respect to s:*

$$\frac{\partial}{\partial s}[-\beta s i p_s] = -\beta i p_s$$

$$\frac{\partial}{\partial s}[-v s p_s] = -v p_s$$

$$\frac{\partial}{\partial s}[+\beta s i p_i] = +\beta i p_i$$

$$\frac{\partial}{\partial s}[all\ other\ terms] = 0$$

*Collecting:*

$$\frac{\partial H}{\partial s} = -\beta i p_s - v p_s + \beta i p_i$$

*Rearranging:*

$$\frac{\partial H}{\partial s} = +\beta i p_i - \beta i p_s - v p_s$$

*Factoring:*

$$\frac{\partial H}{\partial s} = \beta i(p_i - p_s) - p_s v$$

Therefore:

Applying $\dot{p}_s$s = −∂H/∂s:

$$\dot{p}_s = -\frac{\partial H}{\partial s} = -\beta i(p_i - p_s) + p_s v$$

**Step 3 : Find the Co-State Equation for $p_i$**

PMP co-state rule:

$$\dot{p}_i = -\frac{\partial H}{\partial i}$$

*Differentiating H term by term with respect to i:*

$$\frac{\partial}{\partial i}[\lambda_i i] = \lambda_i$$

$$\frac{\partial}{\partial i}[-\beta s i p_s] = -\beta s p_s$$

$$\frac{\partial}{\partial i}[+\beta s i p_i] = +\beta s p_i$$

$$\frac{\partial}{\partial i}[-\gamma i p_i] = -\gamma p_i$$

*Collecting:*

$$\frac{\partial H}{\partial i} = \lambda_i - \beta s p_s + \beta s p_i - \gamma p_i$$

*Rearranging:*

$$\frac{\partial H}{\partial i} = \lambda_i + \beta s p_i - \beta s p_s - \gamma p_i$$

*Factoring:*

$$\frac{\partial H}{\partial i} = \lambda_i + \beta s(p_i - p_s) - \gamma p_i$$

Therefore:

Applying $\dot{p}_i = -\partial H/\partial i$:

$$\dot{p}_i = -\frac{\partial H}{\partial i} = -\lambda_i - \beta s(p_i - p_s) + \gamma p_i$$

**Step 4 : Apply the Transversality Conditions**

The terminal state is free with no terminal cost, so the PMP gives:

$$p_s(T) = 0, \quad p_i(T) = 0$$

**Complete Two-Point Boundary Value Problem (TPBVP) :**

*State equations (forward, t = 0 to 50):*

$$\dot{s} = -\beta s i - v * s, \quad s(0) = 0.95$$

$$i = \beta s i - \gamma i, \quad i(0) = 0.01$$

*Co-state equations (backward, t = 50 to 0):*

$$\dot{p}_s = -\beta i(p_i - p_s) + p_s v*, \quad p_s(50) = 0$$

$$\dot{p}_i = -\lambda_i - \beta s(p_i - p_s) + \gamma p_i, \quad p_i(50) = 0$$

*Optimal control:*

$$v*(t) = \begin{cases} v_M & \text{if } \varphi(t) < 0 \\ 0 & \text{if } \varphi(t) > 0 \end{cases}$$

The state and co-state systems are coupled through v*(t) and solved numerically by AMPL/IPOPT (N = 500 time steps, dt = 0.1).

**Step 5 : Determine the Switching Function φ(t)**

Rewrite H to isolate the v-dependent term:

$$H = (\lambda_v - p_s s)v + \Psi(s, i, p_s, p_i)$$

*Differentiating the v-term with respect to v:*

$$\frac{\partial}{\partial v}[(\lambda_v - p_s s)v] = \lambda_v - p_s s$$

Therefore:

$$\varphi(t) = \frac{\partial H}{\partial v} = \lambda_v - p_s(t) \cdot s(t)$$

**Numerical verification (AMPL/IPOPT, case1_results.csv) :**

At t = 0, substituting $p_s s(0) = 58.038$ and s(0) = 0.95:

$$\varphi(0) = \lambda_v - p_s(0) \cdot s(0)$$
$$= 2.0 - 58.038 \cdot 0.95$$
$$= 2.0 - 55.136$$
$$= -53.136$$

At t = T = 50, using $p_s s(50) = 0$ (transversality):

$$\varphi(50) = \lambda_v - p_s(50) \cdot s(50)$$
$$= 2.0 - (0) \cdot s(50)$$
$$= 2.0 - 0$$
$$= +2.0 > 0$$

**Step 6 : Determine the Optimal Control Policy**

H is linear in v; to minimise H over $v \in [0, v_m]$ choose v by the sign of $\varphi(t)$:

- $\varphi(t) < 0$ : H decreases as v increases → $v^* = v_m = 0.05$ (vaccinate at maximum rate)
- $\varphi(t) > 0$ : H increases as v increases → $v^* = 0$ (no vaccination)
- $\varphi(t) = 0$ on an interval: no singular arc exists in Case I (Theorem 4.8, Della Rossa et al., 2025).

Therefore:

$$v*(t) = \begin{cases} v_M & \text{if } \varphi(t) < 0 \\ 0 & \text{if } \varphi(t) > 0 \end{cases}$$

**Step 7 : Find the Switching Time t***

The SIR co-state equations are coupled and nonlinear, so $\varphi(t) = 0$ has no closed-form solution. The switch is located from the simulation output:

$$\varphi(29.8) = -0.001475 < 0$$

$$\varphi(29.9) = +0.033501 > 0$$

The sign changes between t = 29.8 and t = 29.9.

Therefore:

$$t* \approx 29.9$$

**Step 8 : Write the Final Optimal Control**

Before t* : $\varphi(t) < 0 \rightarrow v^* = v_m = 0.05$. After t* : $\varphi(t) > 0 \rightarrow v^* = 0$.

Therefore:

The optimal bang-bang vaccination control for Case I is:

$$v*(t) = \begin{cases} v_M = 0.05 & \text{for } 0 \le t < t^* \approx 29.9 \\ 0 & \text{for } t^* \approx 29.9 < t \le 50 \end{cases}$$

Without the ICU constraint, i(t) peaks at $i_{max} \approx 0.0412$ at $t \approx 17.9$, exceeding the threshold $i_m$ = 0.04. This motivates the state constraint imposed in Case II.

# Case II: Derivation of the Optimal Vaccination Control with Constraint

*SIR Epidemic Model - With ICU State Constraint*

Della Rossa, Freddi & Goreac (2025). JOTA 204:8. DOI: 10.1007/s10957-024-02598-w

**Problem Statement**

Minimise the cost functional:

$$J(v) = \int_0^T [\lambda_v v(t) + \lambda_i i(t)] \, \mathrm{dt}$$

subject to the SIR state equations:

$$\dot{s} = -\beta s i - v s,$$
$$i = \beta s i - \gamma i,$$

with the additional ICU capacity constraint:

$$g(i) = i(t) - i_M \leq 0$$

Parameters: $\beta = 0.30$, $\gamma = 0.10$, $\lambda_v = 2.0$, $\lambda_i = 15.0$, s(0) = 0.95, i(0) = 0.01, T = 50, 0 $\leq$ v(t) $\leq$ $v_m$ = 0.05, $i_m$ = 0.04.

**Step 1 : Introduce the State Constraint**

The admissible trajectories satisfy:

$$g(i) = i(t) - i_M \leq 0$$

The constraint becomes active when:

$$i(t) = i_M$$

Whether this constraint can be sustained over a time interval, or is touched only instantaneously, is examined in Step 6.

**Step 2 : Form the Augmented Hamiltonian**

For state-constrained optimal control, PMP introduces a nonnegative measure multiplier:

$$d\mu(t) \geq 0$$

associated with the constraint.

The augmented Hamiltonian becomes

$$\bar{H} = H + (i - i_M)d\mu$$

Substituting the Case I Hamiltonian:

$$\bar{H} = \lambda_v v + \lambda_i i - \beta s i p_s - v s p_s + \beta s i p_i - \gamma i p_i + (i - i_M)d\mu$$

**Step 3 : Find the Co-State Equation for $p_s$s**

The co-state equation for $p_s$ is unchanged:

$$\dot{p}_s = -\beta i(p_i - p_s) + p_s v$$

**Step 4 : Find the Co-State Equation for $p_i$**

PMP co-state rule:

$$\dot{p}_i = -\frac{\partial \bar{H}}{\partial i}$$

Differentiate term-by-term:

$$\frac{\partial \bar{H}}{\partial i} = \lambda_i - \beta s p_s + \beta s p_i - \gamma p_i + d\mu$$

Therefore,

$$\dot{p}_i = -\lambda_i + \beta s p_s - \beta s p_i + \gamma p_i - d\mu$$

**Step 5 : Apply the Complementarity Conditions**

The multiplier is a nonnegative measure satisfying:

$$d\mu \geq 0$$

and

$$\int_0^T (i - i_M)d\mu = 0$$

The constraint is inactive if:

$$i(t) < i_M$$

Then,

$$d\mu = 0$$

The constraint is touched when:

$$i(t) = i_M$$

Then the multiplier may be active; in this problem it concentrates at the single contact point rather than over an interval (Della Rossa et al., 2025, Lemma 4.4):

$$d\mu \geq 0$$

and the multiplier dμ acts on the co-state dynamics at the contact; whether the infection can remain on the boundary over an interval is tested in Step 6.

**<u>Step 6 : Test for a Boundary Arc</u>**

When the constraint is active:

$$i(t) = i_M$$

Since $i_M$ is constant,

$$i = 0$$

Substituting the state equation:

$$i = \ \beta si - \gamma i = 0$$

$$i(\beta s - \gamma) = 0$$

$$\beta s - \gamma = 0$$

$$s = \frac{\gamma}{\beta}$$

**<u>Step 7 : Feasibility of Boundary Control</u>**

Differentiate the boundary condition:

$$s = \frac{\gamma}{\beta}$$

Since γ and β are constants,

$$\dot{s} = 0$$

$$-\beta si - vs = 0$$

$$-\beta\left(\frac{\gamma}{\beta}\right)i_M - vs = 0$$

$$-\gamma i_M - vs = 0$$

Since $\gamma$, $i_M$, $v$, $s > 0$, $\dot{s}$ cannot be 0 unless $v < 0$, which violates the control bounds $0 \leq v(t) \leq v_m$.

Since maintaining the boundary would require $v < 0$, no boundary arc is admissible. The ICU constraint is therefore met at a single contact point $t_0 \approx 17.9$ (not a sustained arc), in agreement with Della Rossa et al. (2025), Lemma 4.4 and Theorem 4.8.

**Step 8 : Determine the Optimal Control Policy**

Because the augmented Hamiltonian remains purely linear in the control variable v, the formulation of the switching function $\varphi(t)$ is unaltered:

$$\varphi(t) = \lambda_v - p_s(t) \cdot s(t)$$

The theoretical optimal control therefore retains the bang-bang form (at most one switch):

$$v*(t) = \begin{cases} v_M & \text{if } \varphi(t) < 0 \\ 0 & \text{if } \varphi(t) > 0 \end{cases}$$

The switching function keeps the form $\varphi(t) = \lambda_v - p_s(t)s(t)$, but in the constrained case the co-states carry the state-constraint multiplier $d\mu$. The plain numerical $\varphi$ crossing must therefore be interpreted together with the active ICU constraint.

**Step 9 : Switching-Function Crossing and Contact Point**

The SIR co-state equations are coupled and nonlinear, so $\varphi(t) = 0$ has no closed-form solution. The plain switching function is evaluated from the simulation output:

$$\varphi(16.2) = -0.103424 < 0$$
$$\varphi(16.3) = +0.026385 > 0$$

The plain switching function changes sign between t = 16.2 and t = 16.3.

This plain crossing is denoted:

$$t_\varphi \approx 16.3$$

This plain crossing does not set the constrained control switch. Because the ICU constraint is active around the infection peak, the co-state $p_i$ has a jump at the contact, so the realized optimal control stays at $v = v_m$ through the growth phase up to the contact point $t_0 \approx 17.9$, and is then relaxed.

**Step 10 : Write the Final Optimal Control**

Up to the contact point the trajectory is below the ceiling and $v^* = v_m = 0.05$; after the contact the numerical IPOPT solution relaxes v toward 0, whereas the exact optimal policy remains bang-bang with at most one switch.

Therefore:

The optimal (theoretical bang-bang) vaccination control for Case II is:

$$v*(t) = \begin{cases} v_M = 0.05 & \text{for } 0 \le t < t_0 \approx 17.9 \\ 0 & \text{for } t_0 \approx 17.9 < t \le 50 \end{cases}$$

Theoretically the optimum remains bang-bang with at most one switch (Della Rossa et al., 2025, Theorem 4.8); the gradual decline returned by IPOPT is a discretisation effect, not the exact theoretical control.